Neurocritical Care – Chapter Summary

TBI • Stroke • SAH • Seizures • Neuromuscular • Brain Death

ICU / EDIC Board Review • MCQ-Oriented

1. Intracranial Physiology

- **Monro–Kellie:** brain + blood + CSF = constant volume
- **CPP = MAP − ICP** (target 60–70 in TBI)
- **Normal ICP** 5–15 mmHg; treat if > 22 (BTF 4th ed.)
- **Cerebral autoregulation:** CBF constant at MAP 50–150 (shifted right in chronic HTN)
- **CO_2:** CBF changes ~ 3–4% per 1 mmHg $PaCO_2$
- **$CMRO_2$** ↓ 7% per 1 °C ↓ temperature
- **ICP waveform:** P1 (percussion) > P2 normally; P2 > P1 = ↓ compliance
- **Lundberg A waves** (plateau 50–100 mmHg × 5–20 min) = pathological

2. Raised ICP Management (tiered)

- **Tier 0:** head 30°, neck neutral, sedation/analgesia, normothermia, Na 140–145, $PaCO_2$ 35–40, avoid hypotension & hypoxia
- **Tier 1:** ↑ sedation, intermittent CSF drainage (EVD), **osmotherapy** (mannitol 0.25–1 g/kg or hypertonic saline 3–23.4%)
- **Tier 2:** mild hyperventilation ($PaCO_2$ 32–35), neuromuscular blockade, MAP challenge
- **Tier 3:** barbiturate coma, decompressive craniectomy, mild hypothermia (35–36 °C)

Prophylactic hyperventilation ($PaCO_2$ < 25) & steroids in TBI → harm (CRASH)

- **Mannitol:** check osmolar gap < 20 (renal risk); **HTS:** Na ceiling ~ 155–160

3. Traumatic Brain Injury

- **GCS ≤ 8** = severe → intubate; avoid SBP < 100–110 & SpO_2 < 90%
- **BTF thresholds:** ICP > 22, CPP 60–70, SBP ≥ 100 (50–69 yrs) / ≥ 110 (15–49 & > 70)
- **ICP monitoring:** severe TBI with abnormal CT (or normal CT + ≥ 2 of: age > 40, posturing, SBP < 90)
- **DECRA:** early bifrontal craniectomy → worse outcome
- **RESCUEicp:** craniectomy for refractory ICP → ↓ mortality, ↑ vegetative state
- **Tranexamic acid** within 3 h in mild–moderate TBI (CRASH-3)
- **Seizure prophylaxis:** levetiracetam / phenytoin for 7 days (early seizures only)
- **$PbtO_2$ < 20 mmHg** = brain hypoxia
- **Hb:** liberal transfusion (Hb < 10) may improve outcome (HEMOTION neutral; TRAIN: liberal better in acute brain injury)

4. Acute Ischemic Stroke

- **IV thrombolysis:** alteplase 0.9 mg/kg (10% bolus) or **tenecteplase 0.25 mg/kg** within 4.5 h
- **BP before lysis < 185/110**; after < 180/105 for 24 h
- **Thrombectomy:** LVO up to 24 h (DAWN, DEFUSE-3 – mismatch); large core also benefits (SELECT2, ANGEL-ASPECT)
- No lysis: permissive HTN up to 220/120
- **Malignant MCA infarct:** decompressive hemicraniectomy within 48 h, age < 60 (DESTINY, DECIMAL, HAMLET)
- **Cerebellar infarct:** EVD + suboccipital decompression
- Glucose 140–180; avoid hyperthermia; aspirin 24 h after lysis

5. Intracerebral Hemorrhage

- **ICH score:** GCS, volume ≥ 30 mL, IVH, infratentorial, age ≥ 80
- **BP:** target SBP 140 (INTERACT2/3), avoid < 110–130 drop
- **Reversal:** warfarin → 4F-PCC + vit K 10 mg; dabigatran → idarucizumab; Xa inhibitors → andexanet alfa or PCC
- **Platelet transfusion** on antiplatelet: harm (PATCH)
- **Surgery:** cerebellar > 3 cm / brainstem compression / hydrocephalus; minimally invasive lobar evacuation (ENRICH) beneficial
- **TXA:** ↓ hematoma growth, no outcome benefit (TICH-2)
- **Care bundle** (INTERACT3): BP, glucose, temperature, anticoagulation reversal → better outcome

6. Subarachnoid Hemorrhage

- Thunderclap headache; CT (sensitivity ~100% < 6 h) → LP (xanthochromia) → CTA
- **Grading:** WFNS, Hunt & Hess, modified Fisher (vasospasm risk)
- **Secure aneurysm early** (< 24–72 h); coiling preferred when feasible (ISAT)
- **Before securing:** SBP < 160; TXA short-term only
- **Nimodipine 60 mg q4h × 21 days** (↓ poor outcome, not vasospasm)
- **DCI** day 3–14: euvolemia; induced hypertension; endovascular therapy (angioplasty / intra-arterial vasodilator)
- **TCD:** MCA velocity > 120 cm/s; **Lindegaard ratio > 3** = vasospasm
- **Complications:** hydrocephalus (EVD), rebleeding, hyponatremia (SIADH / cerebral salt wasting), neurogenic stunned myocardium, seizures

Avoid prophylactic hypervolemia / triple-H, statins, magnesium (no benefit)

7. Status Epilepticus

- **Definition:** seizure ≥ 5 min or ≥ 2 without recovery
- **0–5 min:** ABC, glucose, thiamine
- **5–20 min:** benzodiazepine – lorazepam 4 mg IV / midazolam 10 mg IM / diazepam
- **20–40 min:** levetiracetam 60 mg/kg = fosphenytoin 20 PE/kg = valproate 40 mg/kg (ESETT)
- **Refractory (40 min+):** intubate, midazolam / propofol / ketamine / thiopental infusion, continuous EEG
- **Super-refractory:** ≥ 24 h on anesthetics → ketamine, ketogenic diet, immunotherapy (NORSE)
- **NCSE** in coma: requires EEG; consider in unexplained ↓ consciousness
- Avoid valproate in pregnancy, liver disease, mitochondrial disease

8. Guillain–Barré Syndrome

- Ascending symmetric weakness, areflexia, after infection (Campylobacter); CSF albuminocytologic dissociation
- **Variants:** Miller Fisher (anti-GQ1b: ophthalmoplegia, ataxia, areflexia), AMAN (axonal)
- **Intubate:** VC < 20 mL/kg, NIF < 30, MEP < 40, bulbar dysfunction (20/30/40 rule)
- **EGRIS** score predicts ventilation
- **Rx:** IVIG 0.4 g/kg × 5 days = plasma exchange; **no steroids**
- **Dysautonomia:** BP lability, arrhythmias – avoid succinylcholine

9. Myasthenic Crisis

- Triggers: infection, surgery, drugs (aminoglycosides, FQ, Mg, β-blockers, neuromuscular blockers)
- **Rx:** plasma exchange (faster) or IVIG; steroids later (initial worsening); hold pyridostigmine if intubated
- **Cholinergic crisis:** miosis, salivation, bradycardia, fasciculations (edrophonium no longer used)
- Very sensitive to non-depolarizing NMB; resistant to succinylcholine
- Prefer NIV early (BiPAP) if no bulbar weakness / hypercapnia

10. Spinal Cord Injury

- **High cervical (C3–5):** diaphragm loss → ventilation
- **Neurogenic shock:** hypotension + bradycardia (above T6) → NE, atropine
- **MAP 85–90 × 7 days** (weak evidence)
- **Steroids NOT recommended** (NASCIS controversy)
- **Autonomic dysreflexia** (> T6, chronic): severe HTN triggered by bladder / bowel distension
- Succinylcholine safe < 48–72 h, then hyperkalemia risk

11. CNS Infections

- **Bacterial meningitis:** dexamethasone 10 mg q6h with/before first antibiotic dose (pneumococcus benefit)
- **Empiric:** ceftriaxone + vancomycin (+ ampicillin if > 50 yrs / immunocompromised / pregnancy – Listeria)
- **CT before LP** if: focal deficit, papilledema, GCS < 10, seizure, immunocompromised
- **HSV encephalitis:** temporal lobe; CSF PCR; acyclovir 10 mg/kg q8h
- **Autoimmune encephalitis:** anti-NMDAR (psychiatric, dyskinesias, ovarian teratoma) → steroids, IVIG, PLEX

12. Delirium, Sedation & Weakness

- **Screening:** CAM-ICU, ICDSC
- **PADIS:** light sedation, propofol / dexmedetomidine over benzodiazepines
- **Antipsychotics:** no benefit in delirium (MIND-USA, AID-ICU)
- **Prevention:** ABCDEF bundle, early mobility, sleep hygiene
- **ICU-acquired weakness:** MRC sum score < 48; critical illness polyneuropathy vs myopathy
- Risk factors: sepsis, steroids, NMB, hyperglycemia, immobility

13. Brain Death (Death by Neurologic Criteria)

- **Prerequisites:** known irreversible cause; exclude sedation, NMB, hypothermia (< 36 °C), severe metabolic / endocrine disturbance; SBP ≥ 100 / MAP ≥ 60
- **Clinical:** coma, absent brainstem reflexes (pupils, corneal, oculocephalic, oculovestibular, cough, gag), no response to central pain
- **Apnea test:** $PaCO_2$ ≥ 60 and ≥ 20 above baseline, pH < 7.30, no breathing
- **Ancillary tests** (if exam incomplete): 4-vessel angiography, nuclear perfusion scan, TCD, EEG
- Spinal reflexes may persist

14. Other Topics

- **PRES:** HTN, eclampsia, immunosuppressants (tacrolimus) → parieto-occipital edema
- **Cerebral venous thrombosis:** headache, seizures, hemorrhagic infarct → anticoagulation even with hemorrhage
- **Osmotic demyelination:** Na correction > 8–10 mmol/L/24 h
- **Wernicke:** thiamine before glucose
- **Neuroleptic malignant syndrome** vs **serotonin syndrome** (clonus, hyperreflexia, rapid onset)

15. High-Yield MCQ Pearls

- ☑ CPP = MAP − ICP; TBI target 60–70
- ☑ TBI: treat ICP > 22 mmHg
- ☑ Steroids in TBI → harm (CRASH)
- ☑ DECRA worse; RESCUEicp ↓ mortality
- ☑ Lysis BP < 185/110; ICH target SBP 140
- ☑ Nimodipine in SAH ↑ outcome, not vasospasm
- ☑ Lindegaard > 3 = vasospasm
- ☑ ESETT: levetiracetam = fosphenytoin = valproate
- ☑ GBS: IVIG or PLEX, no steroids
- ☑ Bacterial meningitis: dexamethasone before antibiotics
- ☑ Apnea test: $PaCO_2$ ≥ 60 & ↑ 20
- ☑ Antipsychotics don't treat delirium

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